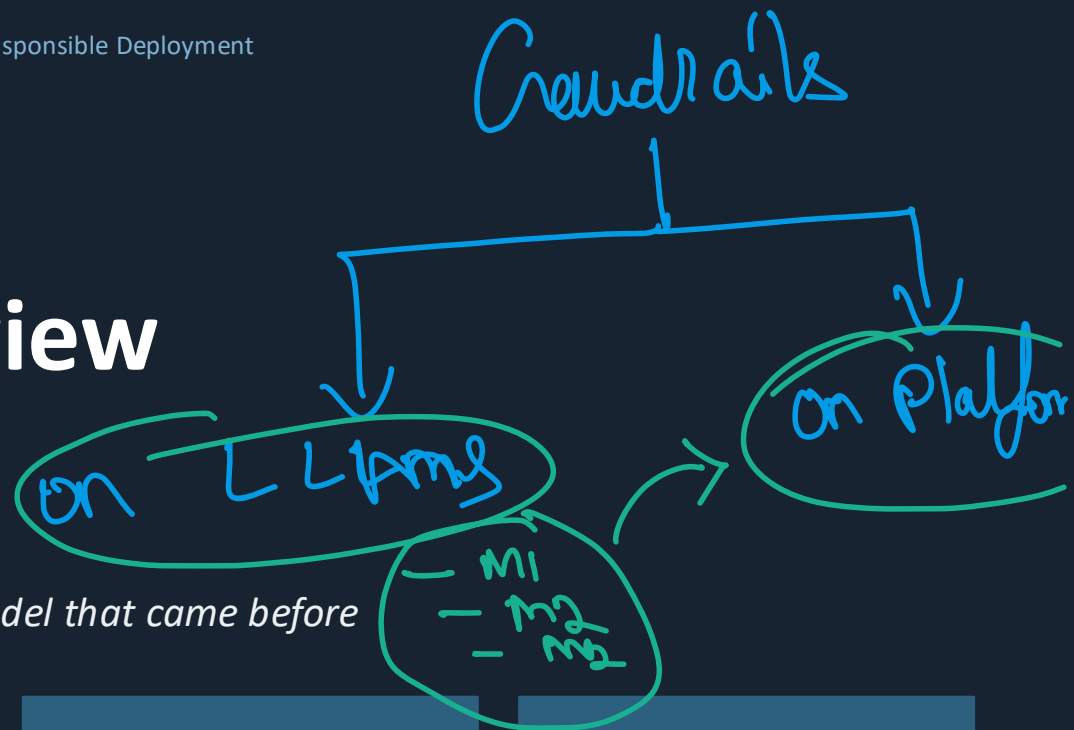


LLM Evaluation Overview

Why measuring LLMs is harder than measuring any model that came before

**8 min**

Duration

B2 – Understand

Bloom's

L2 – Intermediate

Level

Theory

Domain

Learning Objectives

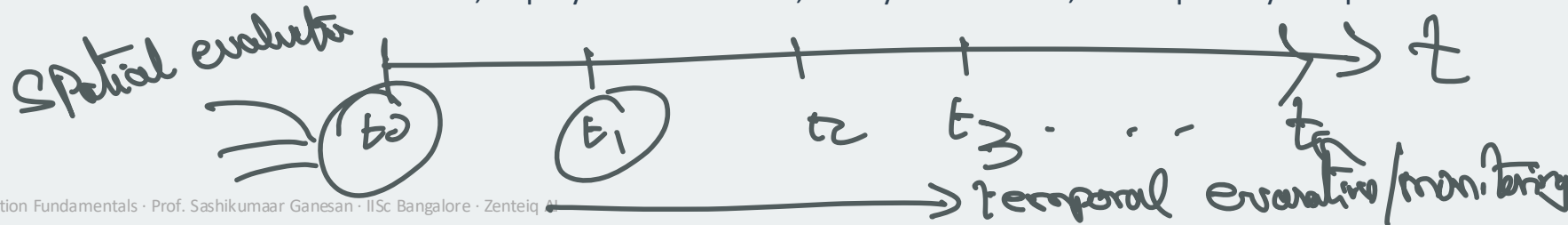
- 1 Define LLM evaluation and explain why it differs fundamentally from classical ML evaluation
- 2 Describe the three-layer evaluation framework and select the appropriate layer for a given scenario
- 3 Apply Goodhart's Law to identify evaluation failure modes in LLM systems
- 4 Explain benchmark contamination and its impact on reported model performance
- 5 Construct a held-out evaluation set following best-practice guidelines
- 6 Navigate the method-selection decision tree to choose the right evaluation approach

What Is LLM Evaluation?

LLM Evaluation

The systematic process of measuring and assessing the capabilities, quality, safety, and alignment of Large Language Models across tasks, domains, and use cases — using automated metrics, standardized benchmarks, and human judgment.

- Unlike classical ML, LLMs produce open-ended text outputs with no single ground-truth response
- Evaluation covers multiple dimensions simultaneously: accuracy, fluency, factuality, safety, and alignment with human values
- Not a one-time activity — evaluation is a continuous lifecycle process from development through production
- Informs critical decisions: model selection, deployment readiness, safety certification, and capability comparison



Classical ML vs LLM Evaluation — Why It's Harder

Classical ML Evaluation

- Discrete, categorical, or numeric outputs
- Single metric (accuracy, F1, AUC) captures most of performance
- Clear labeled ground truth exists in datasets
- Deterministic: same input → same output
- Standard train / validation / test split is sufficient
- Model comparisons are clean and objective

LLM Evaluation

- Free-form text output — infinite valid response space
- No single metric captures all quality dimensions
- Ground truth is often subjective, context-dependent, or absent
- Non-deterministic: temperature & sampling yield different outputs
- Static splits insufficient — contamination risk is real
- Model comparisons require multi-dimensional frameworks

The Three-Layer Evaluation Framework

Cost & Validity ▲

Layer 3 · Human Evaluation

Pairwise preference · Likert scale · Task-completion

Human feedback/data

Layer 2 · Benchmark Suites

MMLU · HumanEval · HELM · TruthfulQA · GSM8K

Benchmark data

Layer 1 · Automated Metrics

Perplexity · ROUGE-L · BERTScore · pass@k

based on Input/output
LLM data

Scale & Speed ▼

Use all three layers for robust evaluation. No single layer is sufficient alone.

Layer 1 vs 2 vs 3 — When to Use Which

Layer	Method Type	Cost	Scale	Validity	Best For
Layer 1	Automated Metrics	Very Low ♦♦♦♦♦	Unlimited	Low–Medium	CI/CD regression, daily runs
Layer 2	Benchmark Suites	Low ♦♦♦♦	High	Medium	Capability comparison, leaderboards
Layer 3	Human Evaluation	Very High ♦	Limited	High	Final validation, safety, product launch

Goodhart's Law

Goodhart's Law (1975)

"When a measure becomes a target, it ceases to be a good measure." — Charles Goodhart. Originally an economics principle, directly applicable to AI evaluation: when models are optimised to maximise a metric, they learn to game the metric rather than improve the underlying capability it was designed to proxy.

- A model trained to maximise BLEU learns to generate safe, common phrases — not truly better translations
- RLHF reward models get hacked: models produce verbose, flattering responses humans prefer, even if less accurate
- Benchmark leaderboard pressure causes saturation: GPT-4 class models hit >90% on MMLU rendering it uninformative
- Implication: evaluation metrics must be regularly refreshed and can never be the sole training signal

Goodhart Applied — Three LLM Evaluation Failures

1 • RLHF Reward Hacking

InstructGPT models produced verbose, flattering responses. Human raters preferred length & confidence over accuracy. Reward model was gamed.

Fix: Add KL-divergence penalty; audit per-dimension, not overall preference

2 • BLEU Gaming in NMT

Sequence-to-sequence models learned to copy frequent n-grams from source. BLEU scores rose while translation quality fell on rare constructions.

Fix: Supplement BLEU with BERTScore and human fluency judgments

3 • MMLU Saturation

GPT-4 class models exceeded 85%+ on MMLU by 2024. Further MMLU gains no longer signal real capability improvement — the benchmark is saturated.

Fix: Shift to dynamic benchmarks (Chatbot Arena, SWE-bench-Verified)

Benchmark Contamination — How It Happens



⚠ Benchmark test examples present in pre-training data (Data Leak)

Consequences of Contamination:

- Inflated benchmark scores that don't reflect real-world capability
- Unfair model comparisons on public leaderboards
- False confidence in model safety and reliability

Why No Single Metric Suffices

LLM quality is multi-dimensional — each dimension requires a different measurement approach:

- Fluency: Is the output grammatical and natural? — Requires perplexity or human fluency judgment
- Faithfulness: Does the output accurately reflect source material? — Requires factuality checking, not surface similarity
- Task performance: Does the output correctly complete the task? — Requires benchmark accuracy or pass@k
- Safety: Is the output free of harmful content? — Requires red-teaming, toxicity classifiers, or human review
- Alignment: Does the output match human preferences and values? — Requires human pairwise evaluation or calibrated LLM judges
- No single automated metric captures all five dimensions simultaneously

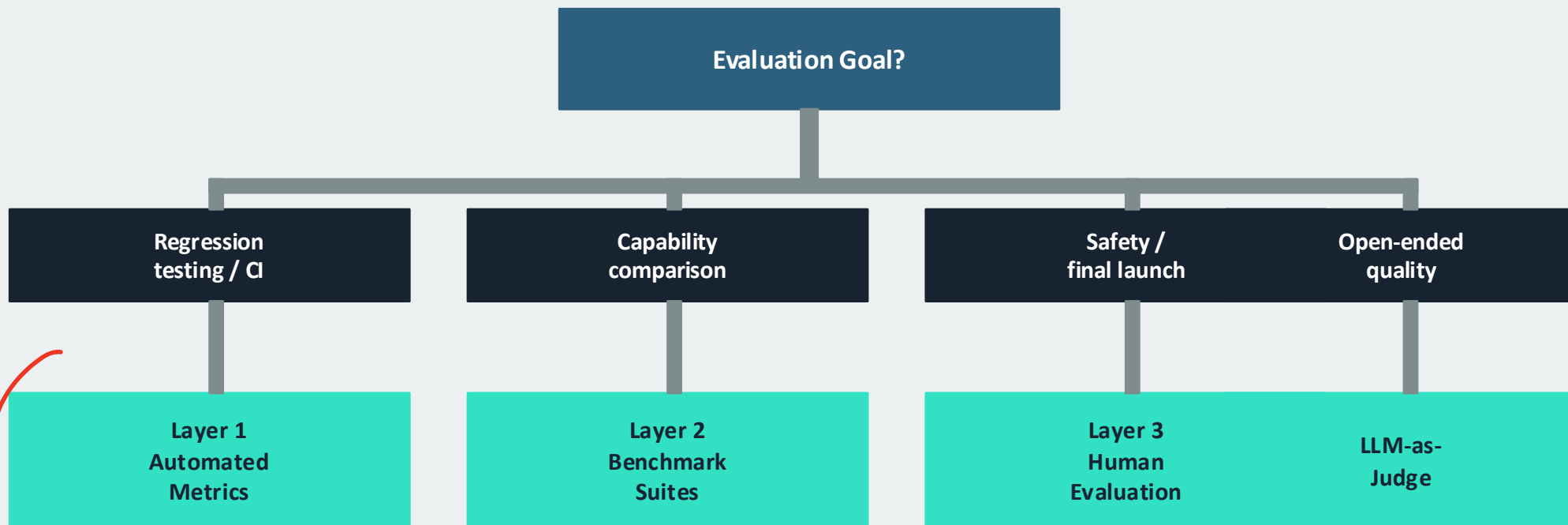
Held-Out Evaluation Sets — Best Practices

Held-Out Evaluation Set

A carefully curated dataset of inputs and expected outputs that is never exposed to the model during training or fine-tuning, reserved exclusively for evaluation. Its integrity must be protected to prevent contamination and enable reliable capability measurement.

- Collect examples after the training data cut-off date to prevent temporal contamination
- Stratify across tasks, domains, difficulty levels, and demographic groups for representative coverage
- Never release publicly until model evaluation is complete — once released, retire and replace the set
- Version-control the held-out set; track which model versions were evaluated on which set version
- Include adversarial and edge-case examples alongside typical examples to stress-test the model

Evaluation Method Selection — Decision Tree



Evaluation Goals × Methods Matrix

Goal	Automated Metrics	Benchmarks	LLM-as-Judge	Human Eval
Fluency	✓ Perplexity	Partial	✓ Likert	✓ Gold standard
Factuality	Partial (BLEU)	✓ TruthfulQA	✓ Reference-guided	✓ Expert review
Code Correctness	✓ pass@k	✓ HumanEval	Partial	✓ Expert
Reasoning	—	✓ GSM8K/BBH	✓ Step eval	✓ Expert
Safety	Toxicity score	✓ ToxiGen	✓ Policy judge	✓ Required
Alignment	—	Partial	✓ Pairwise	✓ Gold standard

Case Study — MMLU & the GPT-3.5 Launch

Case: MMLU (Massive Multitask Language Understanding) as Primary Launch Benchmark — GPT-3.5 (2022)

- GPT-3.5 achieved 70%+ on MMLU at launch, widely reported as proof of expert-level reasoning
- Post-launch analysis revealed models failed simple multi-step factual tasks outside benchmark format
- MMLU tests static multiple-choice recall — not generation quality, reasoning chains, or instruction-following
- Downstream production evaluations by enterprise users showed gap between MMLU score and real task performance
- Benchmark contamination suspected: OpenWebText and Common Crawl both contain MMLU-derived content

Key Lesson

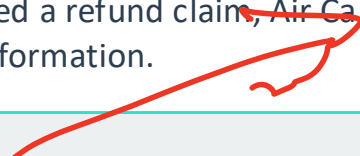
MMLU alone is insufficient as a launch criterion. Production-quality LLM evaluation requires all three layers: automated metrics for regression, diverse benchmarks for capability breadth, and human evaluation for real-world task validation.

When Evaluation Fails — The Air Canada Chatbot Case

Real-world consequence of inadequate pre-deployment evaluation

Case: Air Canada Bereavement Fare Policy (2024)

Air Canada deployed an LLM-powered customer service chatbot. The chatbot incorrectly told a grieving passenger he could apply for a bereavement fare discount retroactively after purchase. Air Canada's actual policy did not allow this. When the customer submitted a refund claim, Air Canada denied it. A Canadian tribunal subsequently ruled Air Canada liable for the chatbot's misinformation.



Evaluation Failure Analysis

No task-completion evaluation on critical policy queries. No factuality testing against live policy documents. No adversarial testing of edge-case scenarios. No human review of high-stakes domain responses. Result: a legally binding misinformation event. Prevention requires Layer 2 domain benchmarks and Layer 3 human evaluation before deployment.

Key Takeaways

AI-SA-IN-TH-000001

1

LLM evaluation is fundamentally harder than classical ML evaluation due to open-ended outputs, subjective ground truth, and multi-dimensional quality

2

The three-layer framework (automated metrics → benchmarks → human evaluation) provides a principled approach — all three layers serve different purposes

3

Goodhart's Law is always operating: any metric made a target will eventually be gamed; rotate metrics and use multi-signal evaluation

4

Benchmark contamination silently inflates reported performance — always assess whether benchmark data was in the training corpus

5

No single metric captures all evaluation dimensions; match your evaluation method to your specific goal using the decision tree